Syllabus - Spring 2026

Note: This syllabus is subject to revision. If I revise the syllabus, then I will provide you with a revised version and announce the changes made.

Class Date and Time

Class Meetings: Section 1 : Mondays and Wednesdays, 1:10–2:25PM
Section 2 : Mondays and Wednesdays, 4:10–5:25PM

On-site Location: Section 1 : 428 Pupin Laboratories
Section 2 : 207 Mathematics Building

About the Professor

Professor: Yaren Bilge Kaya, PhD (she/her)
Lecturer in the Discipline of Industrial Engineering and Operations Research

Office Location: 344 Mudd Hall
E-mail: ybk2105@columbia.edu
(Please begin subject line with "IEOR4004")
Website: www.yarenbilgekaya.com

How should I refer to the professor?
Please refer to me as Prof. Kaya or Dr. Kaya, either option is fine.

About the Teaching Assistants (TAs)

TA(s): Salam Afiouni (she/her)
E-mail: sa4316@columbia.edu
Book a virtual 1-1 time with Salam (Thurs, 10 -11:30 AM)

Tahda Queer (they/them)
E-mail: tq2178@columbia.edu
Book a time with Tahda (Mon, 2:30 - 4 PM)

Andreas Charalampopoulos (he/him)
E-mail: ac5951@columbia.edu
Book a time with Andreas (Wed, 2:30 - 4 PM)

Mohamed Lakhnichi (he/him)
E-mail: ml5000@columbia.edu
Book a time with Mohamed (Fri, 10 - 11:30 AM)

Office Hours

Professor's Office Hours:

What are office hours for?

I look forward to seeing you during office hours. This time is reserved for you as students to use however you want: you may ask questions about the course material, chat about graduate school or career opportunities, discuss challenges you are facing, grab some food I have to share, or just get to know us better.

How are office hours structured?
There will be two formats for office hours: private office hours and group office hours.

Private office hours: During the scheduled office hours listed above, you can reserve a private office hour meeting with each of us in 15-minute increments. You may choose to use this as a 1-1 meeting or you can invite some of your classmates to attend as well (for example, if you all have the same question about a homework problem).

You can sign up for a private office hour time slot using these links:
- Book a private office hour with Dr. Kaya

Group office hours: You do not need to make an appointment for group office hours. Simply join the office hour at either of the Professor's office. You can use these group office hours in the same way you'd use private office hours, to discuss anything you want. The difference is that anyone in the class will be able to listen in. Even if you don't have a specific question to ask, it could be a good idea to join group office hours to learn from the questions your classmates have.

- Dr. Kaya's Group Office Hours: Tuesdays 12:00 PM - 1:30 PM at Mudd 344

If you would like to meet with any of us outside the scheduled office hours, please send us an email to make an appointment.

CA's Office Hours:

Location: Mudd 301 unless otherwise indicated

- Thursdays, 9 -11 AM : Muya & Pierre (mg4907@columbia.edu)
- Thursdays, 11 AM -1 PM : Raphael & Eleni (rrc2149@columbia.edu)
- Thursdays, 1 -3 PM : Katherine & Wenbo (yp2786@columbia.edu)
- Tuesdays, 12 - 2 PM : Hjalmar & Kevin (ksh2166@columbia.edu)

General Course Information	
Course Credits:	3 credit hours
Course Description:	Covers some of the main methods used in IEOR applications involving deterministic models: linear, integer, and nonlinear programming; and network models.
Pre-requisites:	None
Textbook (Optional):	I will make all detailed lecture slides, in-class exercises, case studies, and other course material available for you for free. Most students find that this is enough material for them to learn the course material well. However, if you prefer to use a textbook, the course material can be supplemented with four **optional** textbooks: • Introduction to Linear Optimization. Bertsimas and Tsitsiklis. • Operations Research: Applications and Algorithms. Winston and Goldberg. • Numerical Optimization. Nocedal and Wright. • Introduction to Operations Research. Hillier and Lieberman.
Course Topics:	• Linear Programming: Formulating linear programming models; Graphical solution method; Simplex method; Duality theory; Sensitivity analysis. • Integer Programming: Formulating logical conditions; Branch and bound method; Cutting planes methods; Applications of integer programming. • Network Models: Shortest path, Minimum spanning tree, Maximum flow, Transshipment, Assignment, and Traveling salesperson problems. • Non-linear Programming: Unconstrained optimization; Constrained optimization; Lagrange multipliers; Karush-Kuhn-Tucker conditions; Gradient descent; Newton's method.
Course Outcomes:	At the completion of the course, students should be able to: • Be familiar with various optimization algorithms and techniques. They should understand the strengths and limitations of each algorithm and know when to apply them, • Understand how to identify decision variables, objective functions, and constraints and express them mathematically, • Analyze real-world problems by questioning assumptions and formulate them as optimization models, • Implement optimization models using Gurobi Optimization Software, • Understand the trade-offs between model complexity, computational requirements, and solution quality; and identify limitations and propose improvements for optimization models and algorithms.

Date	Topic	Student Deliverables
21-Jan	Syllabus Review, Intro to Mathematical Modeling	
26-Jan	LP: Intro to Linear Programming (LP)	
28-Jan	LP: Formulation and Graphical Solution	
2-Feb	LP: Introduction to Optimization with Gurobi	**HW1 Available**
4-Feb	LP: The Simplex Method	
9-Feb	LP: Gurobi Large Data Example	**HW2 Available**
11-Feb	LP: Duality and Post-Optimal Analysis	
16-Feb	LP: Case Study	
18-Feb	**Quiz 1 (LP)**	**Project 1 Available**
23-Feb	Intro to Integer Programming (IP)	
25-Feb	IP: Logical Conditions	
2-Mar	IP: B&B and Cutting Planes	
4-Mar	IP: TSP and IP Review	
9-Mar	In-Class Office Hour for Projects	**HW3 Available**
11-Mar	IP: Case Study	
16-Mar	*No Class - Spring Break*	
18-Mar	*No Class - Spring Break*	
23-Mar	Intro to Network Models (NM)	
25-Mar	NM: In Class Game	
30-Mar	NM: Minimum Spanning Tree & Max Flow Problems	
1-Apr	NM: Transportation Models	
6-Apr	**Quiz 2 (IP & NM)**	**Project 2 Available**
8-Apr	Max Flow Problems & Transportation Models	**Project 2 Available**
13-Apr	Case Study | LP & NM (Locational Marginal Pricing)	
15-Apr	Intro to Nonlinear Programming (NLP)	
20-Apr	In-class Office Hour for Projects	
22-Apr	Case Study | Quadratic Programming NLP: Unconstrained Optimization and SciPy	 **HW5 Available**
27-Apr	NLP: Constrained Optimization	
29-Apr	NLP: Algorithmically Solving NLPs	
4-May	End of Semester Question Session & Final Quiz Prep	

Important Dates

Important Dates:

Exams:

Quiz 1: Wednesday, February 18, 2026

Quiz 2: Monday, April 6, 2026

Quiz 3 (Final): Monday, May 11, 2026

No Class:

March 16 & 18, Monday and Wednesday - Spring Break

Please let me know as soon as possible if religious holidays or other circumstances will affect your schedule this term so that we can develop an alternative plan for those times.

Grading

Grading Structure

Student performance will be evaluated using the following weights:

- Assignments (4-6): 15%
- Projects: 45% where each project will correspond to 22.5% of the final grade
- Quizzes: 40% where each quiz will correspond to either 10% or 15% of the final grade.

Note: *the quiz where you score the worst is worth 10%, the remaining two are worth 15%.*

Letter Grade Scale

Letter Grade	Range
A+	100–99%
A	98.9–93%
A-	92.9–90%
B+	89.9–87%
B	86.9–83%
B-	82.9–80%
C+	79.9–77%
C	76.9–73%
C-	72.9–70%
D	69.9–60%
F	Below 59.9%

Assignments and Assessments

Assignments:

All assignments are due **Wednesdays** at **11:59** PM via Canvas as a single PDF file unless otherwise indicated. Homework solutions can be typed or handwritten, but please ensure that any handwritten submissions are legible and converted to PDF.

Please name your submission file: **"UNI_HW#.pdf"**

Also, please make sure your name is listed at the top of the document!

There will be **4-6** assignments. Each of these assignments will be graded out of 100 points. Assignments and their solutions will be distributed via Canvas and Gradescope.

Exams:

What can I use on the exam? What can't I use?

All exams will be closed-book and closed-notes. You may bring one double-sided, handwritten cheat sheet that you prepared yourself. Cheat sheets will be collected with the exam papers.

During the exam, you may not discuss the exam with anyone other than Prof. Kaya, and you may not search the internet or use any unauthorized resources.

What if I can't take the exam during class because I'm in a different time zone, I am not feeling well, or I have something else going on?

If you have any unavoidable conflicts with a scheduled exam, please notify Prof. Kaya as soon as possible so that alternate arrangements can be made.

What if I need accommodations?

Any student requiring special accommodations during the exam should contact Prof. Kaya as soon as possible so that adequate arrangements can be made.

Attendance and Participation Policy:

As this course consists of individuals who are unique and irreplaceable, an important component of this course derives from the insights, experiences, and perspectives you bring to the course. I expect we will learn from each other and invite you to share any thoughts you have regarding how the course material relates to your co-op, personal perspectives, or other experiences.

Do I need to join in during the scheduled class period?

Attendance will not be taken during the scheduled class period because there are many reasons class members may not be able to join during the scheduled class period, including being in different time zones, needing to take time for personal wellness, or other commitments.

Projects:	Students will complete two distinct team projects during the course. • Teams for the first project will be randomly assigned. • For the second project, each student will be placed in a new team (different from the first). Students who underperformed in the first project will be grouped together for the second. • The first project must be completed before beginning the second. Your performance will be evaluated based on three components: 1. The quality of your group's final report 2. Evaluations from your group members 3. Evaluation by the professor or teaching assistants Each project contributes 17.5% to your final grade. The breakdown is as follows: • 20%: Final Report Quality, which should include: – An introduction with a clear problem statement – Data usage and parameter estimation – Methodology and mathematical modeling – Results and conclusions • 5%: Individual Contribution, based on active participation at all stages. – 2.5% will come from in-group peer evaluations – 2.5% will come from your meeting with the TA or professor
AI Policy:	Please refer to each assignment and project for the corresponding AI policy. If no AI policy is present, then the following applies: you **can** use AI-assisted tools, but: 1) you are required to explain in your report how they were used 2) you should have full understanding and knowledge of everything that you submit individually or as a team, including any code, modeling decisions, etc., whether it was produced by you, another member of your team, or the AI 3) you bear the final responsibility for the work you submit. You will not be allowed to use AI tools (or any electronic device) during quizzes. Please note that different classes at Columbia may have different Al policies, and it is the student's responsibility to conform to the expectations for each course.

Flexibility and Feedback	
Course Flexibility:	I am fully committed to making sure that you learn everything you were hoping to learn from this class! I will make whatever accommodations I can to help you finish your problem sets, do well on your projects, and learn and understand the class material. Under ordinary conditions, I am flexible and lenient when students face difficult challenges. After pandemic conditions, that flexibility and leniency is intensified. If you tell me you are having trouble or need more time, I will not judge you or think less of you. I hope you'll extend me the same grace. You **never** owe me personal information about your health (mental or physical). However, you are always welcome to talk to me about things that you're going through if you want. If I can't help you, I usually know somebody who can. If you need extra help, or if you need more time with something, or if you feel like you're behind or not understanding everything, please talk to me. I will work with you. I want you to learn a lot from this class, but I primarily want you to take care of your wellness and well-being.
Homework and Exam Extension Policy:	I recognize that illnesses, deaths in the family, and other traumatic events do occur and I want to help you succeed in this course despite such challenges that might arise. As such, please contact us as soon as possible if you anticipate needing an extension. If you fail to ask for an extension in a timely manner, 25% deductions will be made every day the assignment you failed to to submit on time.
Student Feedback:	I truly welcome your feedback. If there are ways in which this course can be made better for you, I would like to hear your suggestions as soon as possible. Telling us your ideas for improving the course at the end of the semester may help the next time the course is taught, but will not help teach the course this term. Please consider sharing your suggestions early! Additionally, students may provide ongoing feedback on course material using this link.

Academic Honesty

Individual Work Policies:	You are **allowed** (indeed, encouraged) to consult with other students enrolled in the class during the conceptualization of a problem, or ask for technical support in accessing the necessary software. However, all submitted assignments, exams, and projects (including computer codes and outputs, if relevant) should represent your own efforts. In particular, you are not allowed to obtain, look at, use, or in any way attempt to derive advantage from the existence of solutions for this or other classes prepared in prior years, whether these solutions were produced by former students or had been made available by previous or current instructors (including students/instructors from other institutions) or textbook publishers. **By taking this course you agree you will not share the course material.** Prof. Kaya have created most of the course materials themselves. Please honor the time and effort they have put into this course by not sharing the course material to people that aren't in the same course section as you. Among other things, you are not allowed to share any of the course material on online platforms (including but not limited to, Chegg, Reddit, Course Hero) or share material with students taking the course in the future. Academic honesty is fundamental to the activities and principles of the university. All members of the academic community must be confident that each person's work has been responsibly and honorably acquired, developed and presented. Any effort to gain an advantage not given to all students is dishonesty whether or not the effort is successful. The academic community regards academic dishonesty as an extremely serious matter, with serious consequences that range from probation to expulsion. When in doubt about plagiarism, paraphrasing, quoting or collaboration, please consult Prof. Kaya. If there is any indication of cheating the student in question will receive a **zero** for the homework/exam. If the incident results in receiving a zero on a homework assignment, the student will not be allowed to drop his/her lowest homework grade. Additionally, the incident will be reported to the Office Academic Integrity, who will decide the appropriate disciplinary action. More information on Columbia University's Academic Integrity Policy and associated processes can be found **here**.

Other Helpful Resources

Basic Needs Security (Hungry? Or need a place to sleep?):	Any student who has difficulty affording groceries or accessing sufficient food to eat every day, or who lacks a safe and stable place to live, and believes this may affect their performance in the course, is urged to contact Live Well for support. Furthermore, please notify the professors if you are comfortable in doing so. This will enable them to provide any resources that they may possess. Some additional resources that are available: • **Emergency Meal Fund:** Columbia Dining sponsors the Emergency Meal Fund, a program offering 6 meals per term to undergraduate students (CC, SEAS, or GS) in urgent need. The program is intended to provide immediate relief to students who need support but is not a long-term solution to food insecurity. For that reason, a conversation with your school's financial aid or student affairs staff is the recommended first step. Undergraduate CC and SEAS students should contact cc-seas@columbia.edu to schedule a meeting with a staff member in 601 Lerner Hall to request assistance. More information: https://dining.columbia.edu/content/food-insecurity • **The Food Pantry at Columbia and Swipe Out Hunger Program:** Swipe Out Hunger is a national nonprofit that partners with students and universities to provide services and support to help address food insecurity, currently across 74 campuses to date. The Columbia Dining Swipe Out Hunger program will direct funds to the Food Pantry for students facing food insecurity. More information: https://thefoodpantry.studentgroups.columbia.edu/
Changing your name or gender marker:	Columbia University recognizes that some students prefer to identify themselves by a First Name and/or Middle Name other than their Legal Name. For this reason, the University enables students to use a "Preferred Name" where possible in the course of University business and education. More information: https://registrar.columbia.edu/content/faq-preferred-name-policy If you would just like to change the name or gender marker you use for this class, please let us know at any time.

Counseling and Mental Health Resources:	I recognize that students may experience depression, anxiety, family stress, the loss of a loved one, memories of childhood trauma, sexual assault, the loss of important relationships, financial problems, and other major stressors. It is normal for students to seek the service of mental health professionals to provide them with support and skills to cope with these experiences and have a successful academic career during their time in college. Some of the related services available to Columbia University students include: • University Counseling and Psychological Services • The Eating Disorders Team at Columbia • First-Generation Students Team • ULifeline (anonymous & confidential mental health resource) • Gay Health Advocacy Project • Nightline Peer Listening • Columbia University Religious Life • 988 Suicide & Crisis Lifeline
Special Accommodations:	If you have physical, mental, emotional, chronic health, or learning needs that may affect your performance in this course, I encourage you to consider meeting with the Disability Services (DS) to see what accommodations may be helpful for you. The DS helps provide classroom and exam related accommodations and helps ensure course material is accessible to your learning needs. **Note:** The University requires that you provide documentation of your disability to the DS. DS Website: `https://www.health.columbia.edu/content/disability-services` Additionally, I encourage you to meet with me privately to discuss additional adaptations or modifications I can make to the course that might be helpful for you.

Sex & Gender-Based Discrimination Policy

Title IX of the Education Amendments of 1972 protects individuals from sex or gender-based discrimination, including discrimination based on gender-identity, in educational programs and activities that receive federal financial assistance. Columbia's Title IX Policy prohibits sexual harassment, sexual assault, relationship or domestic violence, and stalking. The Title IX Policy applies to the entire community, including students, faculty and staff of all gender identities.

There are a number of things you can do if you or anyone you know has been affected by sex and/or gender-based discrimination.

It's important to know what will happen if you reach out to different people at the university, so you can make an informed decision about how you want to handle the discrimination.

If you are not ready to officially report, but you want to talk to someone:

Some survivors of harassment, discrimination, sexual misconduct or other violations may not be ready or willing to report through a channel that could result in university action. For such individuals, the following **confidential** resources are available:

- Disability Services
- The Ombuds Office: `https://ombuds.columbia.edu/`
- University Counseling and Psychological Services staff
- Healthcare providers

By law, those employees are **not required** to report allegations of sex or gender-based discrimination to the University. However, please note that these options may still have some state or federal obligations to report, for example if someone is in immediate danger of being harmed. Before you begin providing details you want to remain confidential, you are encouraged to ask the person you are talking with to give you an overview of what, if any, information they are required to report if you tell them. This will allow you to make the most informed decision about how much you want to share.

If you are ready to report the violation to the University:

Alleged violations can be reported non-confidentially to:

- The Title IX Coordinator within Student Conduct and Community Standards ("SCCS" or "The Office") by filling this form. Filling out the form allows you to complete the report anonymously (University won't know you are the person who submitted the report) and provide as much or as little of the details you are comfortable sharing at the time of the report.
- Any University employee, including faculty, staff and student employees are **required** by Columbia University to report allegations of sex and gender-based discrimination, including sexual misconduct, to the Title IX Coordinator.
- New York City Police Department (by telephone, 1-212-267-7273)

Talking with your Professors:

I am available to talk with you about anything that is going on in your life, including sex or gender-based discrimination, that you want to share with us. I will treat all discussions as privately as possible, but please remember that as a faculty member, I are considered "Title IX responsible employees" at Columbia and are required to report all allegations of sex or gender-based discrimination to the Title IX Coordinator.

More information: http://www.columbia.edu/cu/studentconduct/documents